

Code explanation (Nr. 10)

Multi-neighbor extension of the diagonal scheme, 3D linear solver, Kurganov

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This document addresses the points raised in last week's call. The goal is to briefly discuss all the ideas to reason on what I will work on before writing the report.

- In Section 1 I work out the math of allowing additional diagonal stencils beyond the single nearest-corner pair. This is the main point I want to continue pursuing.
- In Section 2 I note some thoughts about Mr. Monasse's suggestion that in 3D we could replace the LU decomposition with an iterative solver and only update the locally varying weights. This seems really involved and won't be continued, but might work.
- In Section 3 I note some thoughts about the issues of my Kurganov analysis I did last week. This section is rather for completeness. I won't follow this in more detail for the project.
- In Section 4 I talk about some non-physical edge cases that I used to test the code on and comment on their generalizability.
- In Section 5 I collect the open todos.

The notation throughout follows the last documents: $r = \Delta x / \Delta y$, $\tilde{D}_2 = D_{xy} / 2$ for the $(1, 1)$ stencil, $\tilde{A}_x, \tilde{A}_y$ for the axial coefficients, $\sigma = \text{sign}(D_{xy}) \in \{+1, -1\}$ for the diagonal direction. For our model $\mathcal{M} = \{(1, 0), (1, 1)\}$ we always have $D_{xy} = d_2 \geq 0$, so I work with $\sigma = +1$ and the NE-SW branch throughout.

1 Multi-neighbor extension of the diagonal scheme

Recap from last week (orange).

The 2D diagonal scheme replaces the centered 9-point cross stencil by

$$\mathcal{T}_h^{(1,1)}(U)_{i,j} := \frac{U_{i+1,j+1} - 2U_{i,j} + U_{i-1,j-1}}{\Delta x \Delta y} = r \partial_{xx} U + 2 \partial_{xy} U + r^{-1} \partial_{yy} U + \mathcal{O}(h^2), \quad (1)$$

with $r = \Delta x / \Delta y$, and reads

$$L_h^{\mathcal{D}} U = \tilde{A}_x \partial_{xx}^{(5)} U + \tilde{A}_y \partial_{yy}^{(5)} U + \tilde{D}_2 \mathcal{T}_h^{(1,1)}(U), \quad (2)$$

where $\partial_{xx}^{(5)}, \partial_{yy}^{(5)}$ are the 5-point axial Laplacians. Matching the leading order with $D_{xx} \partial_{xx} U + D_{xy} \partial_{xy} U + D_{yy} \partial_{yy} U$ gives

$$\tilde{D}_2 = \frac{D_{xy}}{2}, \quad \tilde{A}_x = D_{xx} - r \tilde{D}_2, \quad \tilde{A}_y = D_{yy} - \tilde{D}_2 / r. \quad (3)$$

The M-matrix property of the implicit step requires all three coefficients to be non-negative, which is equivalent to

$$\frac{|D_{xy}|}{2D_{yy}} \leq r \leq \frac{2D_{xx}}{|D_{xy}|}. \quad (4)$$

This interval is non-empty for every PSD diffusion tensor, and our physical model is always PSD.

1.1 Why look beyond a single diagonal pair?

The single-pair splitting (3) ties the diagonal weight $\tilde{D}_2$ directly to D_{xy} , and uses the same mesh ratio r in both axial corrections. This limits the r we can choose to a cone, which is dictated by D_{xy} . For systems which are strongly diagonally diffusive, this limits our current method to choosing $r = 1$. This extension relaxes this constraint.

1.2 The general (p, q) -shifted diagonal stencil

For integers $p, q \geq 1$ (the case $p, q \leq -1$ is the anti-diagonal branch and works identically), define

$$\boxed{\mathcal{T}_h^{(p,q)}(U)_{i,j} := \frac{U_{i+p,j+q} - 2U_{i,j} + U_{i-p,j-q}}{p q \Delta x \Delta y}}. \quad (5)$$

The denominator $p q \Delta x \Delta y$ is picked as a normalizer to later cancel out factors after the Taylor expansion. Taylor expansion of $U_{i \pm p, j \pm q}$ around (i, j) to second order gives

$$\begin{aligned} U_{i \pm p, j \pm q} &= U_{i,j} \pm p \Delta x \partial_x U \pm q \Delta y \partial_y U \\ &\quad + \frac{(p \Delta x)^2}{2} \partial_{xx} U + p q \Delta x \Delta y \partial_{xy} U + \frac{(q \Delta y)^2}{2} \partial_{yy} U + \mathcal{O}(h^3). \end{aligned} \quad (6)$$

The first-order terms cancel when we add the two expansions (the sign flips together with the offset). What remains is

$$U_{i+p,j+q} - 2U_{i,j} + U_{i-p,j-q} = (p \Delta x)^2 \partial_{xx} U + 2p q \Delta x \Delta y \partial_{xy} U + (q \Delta y)^2 \partial_{yy} U + \mathcal{O}(h^4). \quad (7)$$

Dividing by $p q \Delta x \Delta y$,

$$\mathcal{T}_h^{(p,q)}(U) = r_{(p,q)} \partial_{xx} U + 2 \partial_{xy} U + r_{(p,q)}^{-1} \partial_{yy} U + \mathcal{O}(h^2), \quad r_{(p,q)} := \frac{p \Delta x}{q \Delta y} = \frac{p}{q} r. \quad (8)$$

The coefficient of the cross term is still exactly 2, independent of (p, q) . The axial coefficients are $r_{(p,q)}$ and $r_{(p,q)}^{-1}$, i.e. the original axial coefficients with r replaced by $(p/q)r$.

Choosing $p > q$ effectively elongates the virtual cell in the x -direction and $p < q$ elongates it in y .

1.3 Why not just replace (1, 1) by a single (p, q) stencil?

My first idea was to use a (p, q) stencil to replace the (1, 1) stencil on a virtual mesh with ratio (p/q)r. I work this out here and show that this only shifts our admissible r-window but does not widen it. This could already be enough for some cases. More notable is what happens if we use multiple stencils at once. Then the method becomes extremely powerfull, but more on that in a bit.

1.3.1 The window of a pure (1, P) stencil.

Take $\mathcal{S} = \{(1, P)\}$ with integer $P \geq 1$, so the virtual mesh ratio is $r_{(1,P)} = r/P$ and its inverse is P/r . The matching equations (19)–(21) (shown later) reduce to a square system in $(\tilde{A}_x, \tilde{A}_y, w)$, where $w := \tilde{D}_2^{(1,P)}$ is the only diagonal weight:

$$(\partial_{xy}): \quad 2w = D_{xy} \implies w = \frac{D_{xy}}{2}, \quad (9)$$

$$(\partial_{xx}): \quad \tilde{A}_x = D_{xx} - r_{(1,P)} w = D_{xx} - \frac{r}{P} \cdot \frac{D_{xy}}{2} = D_{xx} - \frac{r D_{xy}}{2P}, \quad (10)$$

$$(\partial_{yy}): \quad \tilde{A}_y = D_{yy} - r_{(1,P)}^{-1} w = D_{yy} - \frac{P}{r} \cdot \frac{D_{xy}}{2} = D_{yy} - \frac{P D_{xy}}{2r}. \quad (11)$$

Imposing $\tilde{A}_x \geq 0$ and $\tilde{A}_y \geq 0$ (remembering $w = D_{xy}/2 \geq 0$ is automatic) gives the admissible r-window for a pure (1, P) stencil:

$$\boxed{\frac{P D_{xy}}{2 D_{yy}} \leq r \leq \frac{2P D_{xx}}{D_{xy}}}. \quad (12)$$

Comparison with the original (1, 1) window. The single-(1, 1) stencil window from (4) is $[D_{xy}/(2D_{yy}), 2D_{xx}/D_{xy}]$. The single-(1, P) window (12) is the same interval with **both endpoints multiplied by the same factor P**:

$$\left[\frac{D_{xy}}{2D_{yy}}, \frac{2D_{xx}}{D_{xy}} \right] \xrightarrow{(1,1) \rightarrow (1,P)} \left[\frac{P D_{xy}}{2D_{yy}}, \frac{2P D_{xx}}{D_{xy}} \right].$$

The multiplicative width of the window, $(\frac{2D_{xx}}{D_{xy}})/(\frac{D_{xy}}{2D_{yy}}) = \frac{4D_{xx}D_{yy}}{D_{xy}^2}$, is unchanged.

1.3.2 What about a fully general single (P, Q) stencil?

A natural follow-up question is: instead of replacing the (1, 1) stencil with a single (1, P) as in Section 1.3, why not use a fully general (P, Q) stencil with both $P \geq 1$ and $Q \geq 1$? The conclusion will be that a single (P, Q) stencil is in fact equivalent, in terms of admissibility window, to a (1, P/Q) stencil at higher bandwidth cost, and so once we have analyzed the special cases (1, P) and (P, 1) we have already covered the useful single-stencil options at the smallest possible cost.

Window of a pure (P, Q) stencil. Take $\mathcal{S} = \{(P, Q)\}$ with integers $P, Q \geq 1$. The virtual mesh ratio of (8) is $r_{(P,Q)} = (P/Q)r$. The matching equations are the analog of those in Section 1.3, with r/P replaced by $(P/Q)r$:

$$(\partial_{xy}): \quad 2w = D_{xy} \implies w = \frac{D_{xy}}{2}, \quad (13)$$

$$(\partial_{xx}): \quad \tilde{A}_x = D_{xx} - r_{(P,Q)} w = D_{xx} - \frac{P}{Q} r \cdot \frac{D_{xy}}{2} = D_{xx} - \frac{P r D_{xy}}{2Q}, \quad (14)$$

$$(\partial_{yy}): \quad \tilde{A}_y = D_{yy} - r_{(P,Q)}^{-1} w = D_{yy} - \frac{Q}{P r} \cdot \frac{D_{xy}}{2} = D_{yy} - \frac{Q D_{xy}}{2P r}. \quad (15)$$

Imposing $\tilde{A}_x \geq 0$ from (14) gives $r \leq 2Q D_{xx}/(P D_{xy})$. Imposing $\tilde{A}_y \geq 0$ from (15) gives $r \geq Q D_{xy}/(2P D_{yy})$. The constraint $w \geq 0$ is automatic since $D_{xy} \geq 0$. Combining,

$$\boxed{\frac{Q D_{xy}}{2P D_{yy}} \leq r \leq \frac{2Q D_{xx}}{P D_{xy}}.} \quad (16)$$

The width is unchanged. Only the location shifts. The multiplicative width of the window (16) is

$$\frac{2Q D_{xx}/(P D_{xy})}{Q D_{xy}/(2P D_{yy})} = \frac{2Q D_{xx}}{P D_{xy}} \cdot \frac{2P D_{yy}}{Q D_{xy}} = \frac{4 D_{xx} D_{yy}}{D_{xy}^2}, \quad (17)$$

which is identical to the multiplicative width of the original $(1, 1)$ window (4), and identical to the width of the single $(1, P)$ window (12). So the integers P and Q only act on the **location** of the window, through the shift factor Q/P , and not on its width:

$$\left[\frac{D_{xy}}{2D_{yy}}, \frac{2D_{xx}}{D_{xy}} \right] \xrightarrow{(1,1) \rightarrow (P,Q)} \left[\frac{Q}{P} \cdot \frac{D_{xy}}{2D_{yy}}, \frac{Q}{P} \cdot \frac{2D_{xx}}{D_{xy}} \right].$$

Why we do not pursue general (P, Q) further. The two observations above mean that for the **single**-stencil replacement, (P, Q) adds nothing essentially new over $(1, P)$ and $(P, 1)$ where we can pick P such that we reconstruct the same shift (i.e. $P \leftarrow Q/P$ for the $(1, P)$ case). Choosing this simpler model has the advantage that it uses values in our matrix that are closer to the diagonal (which makes our solver faster), while keeping the model mathematically identical. More on the numerical efficiency can be found in Section 1.11.

1.4 Multi-stencil splitting ansatz and its rank

Let $\mathcal{S} \subset \{(p, q) : p, q \in \mathbb{Z}, (p, q) \neq (0, 0)\}$ be a finite set of stencil offsets. The extended ansatz is

$$\boxed{L_h^{\mathcal{D}, \mathcal{S}} U = \tilde{A}_x \partial_{xx}^{(5)} U + \tilde{A}_y \partial_{yy}^{(5)} U + \sum_{(p,q) \in \mathcal{S}} \tilde{D}_2^{(p,q)} \mathcal{T}_h^{(p,q)}(U).} \quad (18)$$

For constant coefficients, replacing each $\mathcal{T}_h^{(p,q)}$ by (8) and matching the leading-order terms with $D_{xx} \partial_{xx} U + D_{xy} \partial_{xy} U + D_{yy} \partial_{yy} U$ gives three matching equations:

$$(\partial_{xy}): \quad 2 \sum_{(p,q) \in \mathcal{S}} \tilde{D}_2^{(p,q)} = D_{xy}, \quad (19)$$

$$(\partial_{xx}): \quad \tilde{A}_x + \sum_{(p,q) \in \mathcal{S}} r_{(p,q)} \tilde{D}_2^{(p,q)} = D_{xx}, \quad (20)$$

$$(\partial_{yy}): \quad \tilde{A}_y + \sum_{(p,q) \in \mathcal{S}} r_{(p,q)}^{-1} \tilde{D}_2^{(p,q)} = D_{yy}. \quad (21)$$

There are $|\mathcal{S}| + 2$ unknowns ($\tilde{A}_x, \tilde{A}_y$, plus one $\tilde{D}_2^{(p,q)}$ per stencil) and exactly 3 matching equations, so the system has $|\mathcal{S}| - 1$ degrees of freedom.

1.5 Two-stencil case 1: $(1, 1)$ plus $(1, P)$ extends the upper r bound

To get a concrete feeling for the extension, take $\mathcal{S} = \{(1, 1), (1, P)\}$ with integer $P \geq 2$. Define $w_1 := \tilde{D}_2^{(1,1)}$, $w_2 := \tilde{D}_2^{(1,P)}$. The matching equations (19)–(21) become

$$w_1 + w_2 = \frac{1}{2} D_{xy}, \quad (22)$$

$$\tilde{A}_x = D_{xx} - r w_1 - \frac{r}{P} w_2, \quad (23)$$

$$\tilde{A}_y = D_{yy} - \frac{1}{r} w_1 - \frac{P}{r} w_2, \quad (24)$$

since $r_{(1,1)} = r$ and $r_{(1,P)} = r/P$, with inverses $1/r$ and P/r respectively. The free parameter is w_2 (then w_1 is fixed by (22)). The four non-negativity constraints are $w_1 \geq 0$, $w_2 \geq 0$, $\tilde{A}_x \geq 0$, $\tilde{A}_y \geq 0$.

Substitution and inequalities. Substituting $w_1 = D_{xy}/2 - w_2$ in (23):

$$\tilde{A}_x = D_{xx} - r\left(\frac{D_{xy}}{2} - w_2\right) - \frac{r}{P} w_2 = D_{xx} - \frac{rD_{xy}}{2} + r\left(1 - \frac{1}{P}\right)w_2.$$

For $P > 1$, the coefficient of w_2 is positive, so $\tilde{A}_x$ increases with w_2 . The constraint $\tilde{A}_x \geq 0$ becomes

$$w_2 \geq \frac{rD_{xy}/2 - D_{xx}}{r(1 - 1/P)} = \frac{P(rD_{xy}/2 - D_{xx})}{r(P - 1)}. \quad (25)$$

Substituting in (24):

$$\tilde{A}_y = D_{yy} - \frac{1}{r}\left(\frac{D_{xy}}{2} - w_2\right) - \frac{P}{r} w_2 = D_{yy} - \frac{D_{xy}}{2r} + \frac{1}{r}(1 - P)w_2.$$

For $P > 1$, the coefficient of w_2 is negative, so $\tilde{A}_y$ decreases with w_2 . The constraint $\tilde{A}_y \geq 0$ becomes

$$w_2 \leq \frac{rD_{yy} - D_{xy}/2}{(P - 1)/r} \cdot \frac{1}{r} = \frac{rD_{yy} - D_{xy}/2}{P - 1}. \quad (26)$$

The two remaining constraints $w_2 \geq 0$ and $w_2 \leq D_{xy}/2$ (the latter from $w_1 \geq 0$) just bound the box. Combining all four,

$$\boxed{\max\left(0, \frac{P(rD_{xy}/2 - D_{xx})}{r(P - 1)}\right) \leq w_2 \leq \min\left(\frac{D_{xy}}{2}, \frac{rD_{yy} - D_{xy}/2}{P - 1}\right)}. \quad (27)$$

When is this feasible? We need the lower bound to be $\leq$ the upper bound. The lower bound is zero (and the upper is positive) exactly when

- $rD_{xy}/2 \leq D_{xx}$, i.e. $r \leq 2D_{xx}/D_{xy}$, the original upper r -bound, in which case the floor at 0 is active.
- $rD_{yy} \geq D_{xy}/2$, i.e. $r \geq D_{xy}/(2D_{yy})$, the original lower r -bound, in which case the upper bound is non-negative.

Inside the original window, $w_2 = 0$ is admissible and the splitting reduces to the single-pair one. The interesting regime is $r > 2D_{xx}/D_{xy}$, i.e. outside the original upper bound. In that regime the lower bound (25) is strictly positive, so $w_2 > 0$ is required. Feasibility requires

$$\frac{P(rD_{xy}/2 - D_{xx})}{r(P - 1)} \leq \frac{D_{xy}}{2} \quad (\text{from } w_2 \leq w_1 + w_2 = D_{xy}/2). \quad (28)$$

$$\frac{P(rD_{xy}/2 - D_{xx})}{r(P - 1)} \leq \frac{rD_{yy} - D_{xy}/2}{P - 1} \quad (\text{from (26)}). \quad (29)$$

Constraint (28): multiply by $2r(P - 1)/P > 0$:

$$2(rD_{xy}/2 - D_{xx}) \leq rD_{xy}(P - 1)/P \iff rD_{xy} - 2D_{xx} \leq rD_{xy}(P - 1)/P \iff rD_{xy}/P \leq 2D_{xx}.$$

So

$$\boxed{\frac{D_{xy}}{2D_{yy}} \leq r \leq \frac{2PD_{xx}}{D_{xy}}}. \quad (30)$$

This is exactly a factor P larger than the original upper bound $r \leq 2D_{xx}/D_{xy}$.

Constraint (29): multiply by $r(P - 1)/P > 0$ (assuming $P > 1$):

$$rD_{xy}/2 - D_{xx} \leq \frac{r}{P}(rD_{yy} - D_{xy}/2) \iff P(rD_{xy}/2 - D_{xx}) \leq r(rD_{yy} - D_{xy}/2),$$

$$\begin{aligned}
&\Longleftrightarrow PrD_{xy}/2 - PD_{xx} \leq r^2D_{yy} - rD_{xy}/2, \\
&\Longleftrightarrow r^2D_{yy} - rD_{xy}\frac{P+1}{2} + PD_{xx} \geq 0.
\end{aligned} \tag{31}$$

This is a quadratic in r with positive leading coefficient $D_{yy} > 0$, so it is ≥ 0 either everywhere (if the discriminant is non-positive) or outside the two roots. The discriminant is

$$\Delta_r = \left(\frac{P+1}{2}\right)^2 D_{xy}^2 - 4PD_{xx}D_{yy}. \tag{32}$$

Sub-case PSD with margin. If $D_{xy}^2 \leq 16PD_{xx}D_{yy}/(P+1)^2$, then $\Delta_r \leq 0$, (31) is automatic, and the only binding constraint is (30). For $P = 1$ this reproduces the PSD condition $D_{xy}^2 \leq 4D_{xx}D_{yy}$, as it should. For $P = 2$ it reads $D_{xy}^2 \leq 32D_{xx}D_{yy}/9$, which is slightly stricter than PSD (a factor $8/9 \approx 0.89$). So as P grows, the extra constraint (31) starts to bite for tensors close to the PSD boundary.

Sub-case near-singular. If $\Delta_r > 0$, then (31) excludes a finite interval of r between the two roots:

$$r_- \leq r \leq r_+, \quad r_{\pm} = \frac{1}{2D_{yy}} \left(\frac{P+1}{2} D_{xy} \pm \sqrt{\Delta_r} \right), \tag{33}$$

is the forbidden zone. In this regime our two-stencil splitting does not work, even though it would work for a smaller P .

So we need to be a bit careful here and acknowledge that the bounds on r do not grow for every case.

1.6 Two-stencil case 2: $(1, 1)$ plus $(P, 1)$ extends the lower r bound

By the same calculation with $r_{(P,1)} = Pr$ (instead of r/P), the roles of $\tilde{A}_x$ and $\tilde{A}_y$ swap and the lower r -bound extends:

$$\boxed{\frac{2D_{xx}}{D_{xy}} \geq r \geq \frac{D_{xy}}{2PD_{yy}}}, \tag{34}$$

again a factor P . The discriminant condition that controls feasibility near the PSD boundary is identical to (32) by symmetry ($D_{xx} \leftrightarrow D_{yy}$). For our model, $D_{xx} \geq D_{yy}$ (since $D_{xx} = (d_1 + d_2)/2$ and $D_{yy} = d_2/2$), so the upper bound is the harder one to extend, and (30) is the more useful direction.

1.7 Three-stencil case

Taking $\mathcal{S} = \{(1, 1), (1, P), (P, 1)\}$ with weights $w_1, w_2, w_3 \geq 0$, the matching equations are

$$\begin{aligned}
w_1 + w_2 + w_3 &= D_{xy}/2, \\
\tilde{A}_x &= D_{xx} - rw_1 - (r/P)w_2 - (Pr)w_3, \\
\tilde{A}_y &= D_{yy} - w_1/r - (P/r)w_2 - w_3/(Pr).
\end{aligned}$$

With two free parameters (e.g. w_2 and w_3). I worked this out on paper, but one can very similarly as before show that the r -window becomes

$$\boxed{\frac{D_{xy}}{2PD_{yy}} \leq r \leq \frac{2PD_{xx}}{D_{xy}}}, \tag{35}$$

again under the constraint that the corresponding discriminant in the analog of (32) stays non-positive. So three stencils extend both bounds by the same factor P ; the multiplicative width of the window is multiplied by P^2 .

1.8 Impact on the geometric (adaptive) grid

A natural follow-up question is what the multi-stencil widening does to the geometric (adaptive) grid that we discussed in call 9.

Repetition from other calls, nothing new here. Just for completeness here.

Geometric grid setup (recap from call 9). We considered a grid with common growth factor $1 + \alpha$ in both directions,

$$\Delta x_i = \Delta x_0 (1 + \alpha)^i, \quad \Delta y_j = \Delta y_0 (1 + \alpha)^j, \quad i \in \{0, \dots, N_x - 1\}, \quad j \in \{0, \dots, N_y - 1\}. \quad (36)$$

This gives the cell-local mesh ratio

$$\tilde{r}_{ij} = \frac{\Delta x_i}{\Delta y_j} = \frac{\Delta x_0}{\Delta y_0} (1 + \alpha)^{i-j} = r_0 (1 + \alpha)^{i-j}, \quad r_0 := \frac{\Delta x_0}{\Delta y_0}. \quad (37)$$

The free parameters are the global growth rate α and the initial aspect ratio r_0 . The grid covers cells with $(i - j)$ ranging from $-(N_y - 1)$ (when $i = 0, j = N_y - 1$) to $+(N_x - 1)$ (when $i = N_x - 1, j = 0$).

Single-(1, 1) scheme: the old bound on α . For our model $\mathcal{M} = \{(1, 0), (1, 1)\}$ the per-cell admissibility (4) specializes to

$$1 \leq \tilde{r}_{ij} \leq 1 + \rho_i, \quad \rho_i := \frac{d_1(x_i)}{d_2(x_i)}. \quad (38)$$

Let $\rho_{\min} := \inf_i \rho_i$ be the worst-case (smallest) ratio across the grid; this is a fixed constant determined by the physical model. Substituting (37) into (38) and asking that the inequality hold at every cell yields two binding global inequalities:

- **Lower bound at the cell with most negative $i - j$.** For $\alpha > 0$, the product $r_0(1 + \alpha)^{i-j}$ is smallest when $i - j$ is most negative, i.e. at the cell $(i, j) = (0, N_y - 1)$ where $i - j = -(N_y - 1)$. The lower bound $\tilde{r}_{ij} \geq 1$ becomes

$$r_0 (1 + \alpha)^{-(N_y - 1)} \geq 1 \iff r_0 \geq (1 + \alpha)^{N_y - 1}.$$

- **Upper bound at the cell with most positive $i - j$.** The product is largest at $(i, j) = (N_x - 1, 0)$ where $i - j = N_x - 1$. Using $1 + \rho_i \geq 1 + \rho_{\min}$, the upper bound $\tilde{r}_{ij} \leq 1 + \rho_i$ becomes

$$r_0 (1 + \alpha)^{N_x - 1} \leq 1 + \rho_{\min} \iff r_0 \leq \frac{1 + \rho_{\min}}{(1 + \alpha)^{N_x - 1}}.$$

Combining the two,

$$(1 + \alpha)^{N_y - 1} \leq r_0 \leq \frac{1 + \rho_{\min}}{(1 + \alpha)^{N_x - 1}}.$$

A feasible r_0 exists if and only if the left side does not exceed the right side, which gives

$$(1 + \alpha)^{N_x + N_y - 2} \leq 1 + \rho_{\min} \iff \boxed{\alpha \leq \alpha_{\max}^{(1,1)} := (1 + \rho_{\min})^{1/(N_x + N_y - 2)} - 1.} \quad (39)$$

Last call I already showed this bound (slightly different notation). We clearly see that this bound is very restrictive and will make this extension almost unusable.

Three-stencil scheme: the new bound on α . With the three-stencil set $\mathcal{S} = \{(1, 1), (1, P), (P, 1)\}$, the admissible r -window (35) specializes for our model to

$$\frac{1}{P} \leq \tilde{r}_{ij} \leq P(1 + \rho_i), \quad (40)$$

since for our tensor $D_{xy}/(2P D_{yy}) = d_2/(2P \cdot d_2/2) = 1/P$ and $2P D_{xx}/D_{xy} = 2P(d_1 + d_2)/(2d_2) = P(1 + \rho_i)$. Both endpoints of the per-cell window are multiplied by P relative to (38). The same argument as above now gives:

- Lower bound: $r_0(1 + \alpha)^{-(N_y-1)} \geq 1/P$, i.e. $r_0 \geq (1 + \alpha)^{N_y-1}/P$.
- Upper bound: $r_0(1 + \alpha)^{N_x-1} \leq P(1 + \rho_{\min})$, i.e. $r_0 \leq P(1 + \rho_{\min})/(1 + \alpha)^{N_x-1}$.

Combining,

$$\frac{(1 + \alpha)^{N_y-1}}{P} \leq r_0 \leq \frac{P(1 + \rho_{\min})}{(1 + \alpha)^{N_x-1}},$$

so a feasible r_0 exists iff

$$(1 + \alpha)^{N_x+N_y-2} \leq P^2(1 + \rho_{\min}) \iff \boxed{\alpha \leq \alpha_{\max}^{(P)} := (P^2(1 + \rho_{\min}))^{1/(N_x+N_y-2)} - 1.} \quad (41)$$

Compared with (39), the factor under the radical has been multiplied by P^2 . This seems like a very strong improvement, but I did not show that numerically yet. It could also be interesting to compare what we gain from using the geometric mesh compared to what the extra cost of having 3 diagonals stencils would be, this is not clear yet and should probably be benchmarked at some point.

1.9 Loss of uniqueness

For $|\mathcal{S}| \geq 2$ the splitting is not unique: there is a $(|\mathcal{S}| - 1)$ -dimensional family of admissible weight vectors $(\tilde{A}_x, \tilde{A}_y, \{\tilde{D}_2^{(p,q)}\})$. In the two-stencil case of Section 1.5 this is the free parameter w_2 , which ranges over the box (27).

Why this matters in practice. The local truncation error and stability constants of the scheme do depend on the chosen w_2 . For instance, $\tilde{A}_x$ and $\tilde{A}_y$ enter the Fourier symbol (call-9 doc, eq. 6) of the discrete diffusion operator, and hence the stability radius and the diagonal dominance of the implicit-step matrix. Choosing w_2 at the boundary of the admissible box (i.e. saturating one of the non-negativity constraints) makes some weights vanish, which weakens the diagonal dominance. Choosing w_2 in the interior strengthens it.

One way to fix uniqueness: a centering criterion. A natural choice is to pick w_2 that maximizes some interior measure of safety, for example

$$w_2^* = \arg \max_{w_2 \in [\text{lower}, \text{upper}]} \min(\tilde{A}_x, \tilde{A}_y, w_1, w_2),$$

This is the value of w_2 that stays as far as possible from all four walls and is cheap to compute.

1.10 Brief note on dynamic and zone-based selection

In the call the idea was raised of choosing the stencil set $\mathcal{S}$ adaptively, either per cell or per zone of the grid. The simplest version would be: pre-compute, in regions where the local mesh ratio sits close to the boundary of (4), add extra $(1, P)$ or $(P, 1)$ stencils; elsewhere use just $(1, 1)$. This keeps the matrix sparse in most of the domain and only widens the stencil where it is structurally necessary.

The complication is that a spatially varying $\mathcal{S}$ hurts solver structure (the row pattern is no longer the same across the grid), and it also requires treating the interface between zones, where neighboring cells have different stencil sets. **I will not investigate this further, but it could be an interesting point for the future works section.**

1.11 Numerical cost/efficiency

The window-widening results of Sections 1.5 through 1.7 come at a real implementation cost.

Each stencil contributes a fixed number of off-diagonal nonzero entries in the implicit-step matrix $A = I - \Delta t L_h^{\mathcal{D}, \mathcal{S}}$:

- The two 5-point axial Laplacians $\partial_{xx}^{(5)}, \partial_{yy}^{(5)}$ contribute 4 off-diagonal entries (east, west, north, south), plus they modify the diagonal.
- Each $\mathcal{T}_h^{(p,q)}$ contributes 2 off-diagonal entries (the NE-SW or NW-SE pair at offset $\pm(p, q)$), and modifies the diagonal.

So the single-(1, 1) scheme has $1 + 4 + 2 = 7$ nonzeros per row, the two-stencil $\{(1, 1), (1, P)\}$ scheme has 9 nonzeros per row, and the three-stencil $\{(1, 1), (1, P), (P, 1)\}$ scheme has 11. In all cases the count is $\mathcal{O}(1)$ per row, so the matrix has $\mathcal{O}(N^2)$ total nonzeros on an $N \times N$ grid and remains sparse. Thus, the non-zero values are not a problem for us.

The bigger issue is the cost of bandwidth in our matrix. Higher values of P increase our bandwidth, which makes our solver slower. This is because the original stencil (1, 1) produced an almost diagonal matrix with values sitting close to the diagonal (low bandwidth). High values of P let us act on cells farther away. Hence, the stencil widens. This makes our LU solvers slower.

1.12 Implementation

I got the neighborhood-method to run. I have not yet produced nice plots, but will make sure to include one in the final report.

2 Iterative solvers for the 3D implicit step

In last week's Section 3 I gave the algebraic 3D extension of the diagonal scheme. Each row of the implicit-step matrix has 13 non-zero entries, and the matrix size on a 500^3 grid sits at the boundary of what is realistic to assemble and factor with a direct sparse solver. In the call Mr. Monasse suggested moving to iterative solvers and only updating the locally varying weight vectors instead of re-assembling the matrix each time step. I will not follow through on that idea, but I just quickly noted some thoughts here.

2.1 Iterative solvers as an alternative

For our 3D matrix the natural alternatives are iterative methods (CG if A is symmetric positive definite, BiCGSTAB otherwise), possibly some kind of preconditioner. Iterative solvers would work well because of:

1. **Symmetry of A .** For our IMEX scheme, $L_h^{\mathcal{D},3D}$ is symmetric in the constant-coefficient case (every stencil is centered), and remains symmetric for variable coefficients if we are careful with how we discretize the products $\tilde{A}_a U$ etc. (using a flux-form discretization with shared half-step coefficients). Then $A = I - \Delta t L_h^{\mathcal{D},3D}$ is symmetric, and the negative-definiteness of $L_h^{\mathcal{D},3D}$ (from the Fourier-symbol argument of last week) implies that A is SPD. CG is then directly applicable.
2. **Good initial guess from the previous time step.** The solution U^{n+1} is generally close to U^n , especially when the time step is moderate. Using U^n as the initial guess for the linear solve typically gives convergence in a handful of iterations, even without a sophisticated preconditioner.

2.2 Mr. Monasse's suggestion: update only the weight vectors

The way I understood the suggestion: instead of treating A as a matrix object that needs to be reconstructed, we treat the implicit step as a function $\text{apply}(v) \mapsto A v$ whose only state is the small set of weight arrays $\{\tilde{A}_x, \tilde{A}_y, \tilde{A}_z, \tilde{D}_{12}, \tilde{D}_{13}, \tilde{D}_{23}\}$. At each time step, only these six 3D arrays are recomputed from the current $C_{10}, C_{11}, \dots$ field; the linear solver itself never sees the matrix.

2.3 Future work

I will not continue working on this part, as it does not really fit into the storyline of the report, and time is running out. Nevertheless, we will probably note it as a "future work", because it would make for a good project of similar scope to what we did so far.

3 Revisiting the Kurganov negativity argument

This section revisits the document of the last call in Section 10. Several of the arguments there were either too strongly stated or rested on assumptions that were not made explicit. I clean them up here.

3.1 The nonlinearity caveat

In last week’s document I treated the KT scheme as a linear discretization and showed that the cross-derivative piece of the divergence reduces to the standard centered 9-point stencil. From this, I concluded the scheme is not positivity-preserving. **This conclusion was stated too strongly: the KT scheme is genuinely nonlinear, and linear analysis captures only part of the story.**

Where the linear analysis is still valid. In regions where the discrete solution is smooth enough that the minmod limiter is inactive, the cell-centered slopes reduce to the centered finite difference, $(u_y)_{j,k} = (u_{j,k+1} - u_{j,k-1})/(2\Delta y)$. In these regions, and with frozen coefficients D_{xx}, D_{xy}, D_{yy} , the diffusive flux divergence is exactly linear in u and the call-9 argument applies verbatim: the cross piece of $(\nabla \cdot P)_{j,k}$ is the centered 9-point stencil with the corner-sign defect.

The corrected statement is: **in any region of the domain and any time interval over which the minmod limiter is inactive everywhere on the cross-stencil footprint of cell (j,k) , KT’s diffusion update at (j,k) inherits the M-matrix violation of the 9-point stencil, and hence does not generically preserve positivity at the corner neighbors.** In regions where the limiter is active, this argument does not apply, and indeed the limiter can locally restore positivity by zeroing out the tangential slopes that drive the cross coupling. **To summarize, the argument is somewhat useless, as the interesting part of KT was exactly the minmod limiter, not the other region.**

3.2 KT fails in the first few steps

An observation from re-examining last week’s runs: every KT positivity violation we saw occurred within the first few percent of the integration interval. After a few steps the negativity vanished and $\min(U)$ recovered to (essentially) machine zero and stayed there for the rest of the run.

This is the opposite of what the smooth-IC hypothesis predicts. The step IC heavily activates the minmod limiter from $t = 0$ onward (every cell next to the discontinuity has its tangential slope clipped), yet the negativity appears. The Gaussian IC barely activates the limiter anywhere, yet there is no negativity. So the recovery cannot be explained by “the limiter eventually catches up”; the limiter is already engaged in the case that does fail.

I really can’t explain this phenomenon, so I will just leave it unanswered for now and point out in the report that this can happen.

3.3 Why our scheme is better than KT

Our scheme treats diffusion implicitly with the diagonal splitting. The implicit step matrix $A = I - \Delta t L_h^D$ has the M-matrix property by construction (Section 1 of last week’s doc), so $A^{-1} \geq 0$ entry-wise and positivity is preserved for any $\Delta t > 0$. The trade-off is the cost of a linear solve per step. The advantages over KT are then:

- Positivity for any time step proven (did not find a similar statement for KT in literature).
- KT has $\Delta t \lesssim h^2/\|D\|$ (which can be a real cost on fine grids), whereas our scheme only inherits the advective CFL $\Delta t(|F_x|/\Delta x + |F_y|/\Delta y) \leq \sqrt{3}$ via the explicit advection sub-step.

- Simpler diffusion discretization (no limiter inside diffusion, no tangential-slope reconstruction).

3.4 Literature about KT and positivity for cross-diffusion

I spent some time searching the literature to check whether the KT scheme is known to preserve positivity for the parabolic problem with a non-zero cross-derivative diffusion term that we use here. I did not find any positivity proof for KT applied to the two-dimensional convection-diffusion problem with $D_{xy} \neq 0$. (neither proving that it is pos., nor neg.).

4 Stress tests on nonphysical tensors

A natural way to dramatize the advantage of the diagonal scheme over the 9-point IMEX scheme is to push the 9-point scheme into a regime where it blows up, while the diagonal scheme does not. I did that in the past, but I think this was a mistake.

4.1 What “nonphysical” means here

By the call-8 argument the model produces tensors satisfying $D_{xy}^2 \leq 4D_{xx}D_{yy}$, i.e. $\mathbf{D} \succeq 0$, by the sum-of-squares argument with $d_\mu \geq 0$. So the natural way to break PSD in a controlled experiment is to artificially set

$$D_{xy} := \eta \sqrt{4D_{xx}D_{yy}}, \quad \eta > 1, \quad (42)$$

with η slightly above 1 (e.g. $\eta = 1.1, 1.5, 2$) so that the tensor has one negative eigenvalue $\lambda_- < 0$. The advection coefficients can be left at their physical values (and indeed should be, since the symmetry of the diffusion tensor is what is being broken, not the drift).

4.2 Behavior of the 9-point scheme on non-PSD tensors

For $\eta > 1$ the symbol of the 9-point operator (call-8 eq. (58)) is

$$\widehat{L}_h^{(9)}(\xi, \eta_{\text{freq}}) = -\frac{4D_{xx}}{h^2} \sin^2 \frac{\xi}{2} - \frac{4D_{yy}}{h^2} \sin^2 \frac{\eta_{\text{freq}}}{2} - \frac{D_{xy}}{h^2} \sin \xi \sin \eta_{\text{freq}},$$

where I write η_{freq} for the frequency to distinguish it from η in (42). The third term has indefinite sign; the call-8 PSD argument shows the full sum is ≤ 0 everywhere iff $D_{xy}^2 \leq 4D_{xx}D_{yy}$. For $\eta > 1$ this fails, and there exist Fourier modes $(\xi, \eta_{\text{freq}})$ (specifically along the NW-SE diagonal $\xi = -\eta_{\text{freq}}$) on which the symbol is **positive**.

4.3 Behavior of the diagonal scheme on non-PSD tensors

The diagonal scheme is, by construction, defined only when the splitting (3) gives non-negative coefficients $\tilde{A}_x, \tilde{A}_y, \tilde{D}_2$. By the call-8 argument the existence of an admissible r is equivalent to PSD of $\mathbf{D}$. For $\eta > 1$ this fails.

In the past I created examples where only the 9-point stencil blew up by choosing specific D_{xy} . These must have been special cases or maybe our blow-up with the diagonal solver was just delayed (or smoothened out by some reason). I will thus not include anything like that in the final report. Instead I will just focus on showing that the 9-point stencil can indeed give us negative concentrations.

5 Open todos / Roadmap to finish project

- Implement the geometric grid for the diagonal solver (still pending from last week).
- Implement the improved output-frequency proposed by Mr. Monasse, based on the relative time-rate of change of the concentrations.
- Compare Becker–Döring and the diagonal solver (if I get this in time from the others).
- The neighboring stencil method is implemented. Now I just need to find interesting cases to show it’s capabilities.
- Write report and create poster.